

Solve the following system:

$$\begin{cases} \log_3(x) - \log_3(y) = 1 \\ x^2 - y^2 = 72 \end{cases}$$

$$\log_3(x) - \log_3(y) = 1 \Rightarrow \log_3\left(\frac{x}{y}\right) = 1$$

$$\frac{\ln\left(\frac{x}{y}\right)}{\ln 3} = 1$$

$$\Rightarrow \ln\left(\frac{x}{y}\right) = \ln 3$$

$$e^{\ln\left(\frac{x}{y}\right)} = e^{\ln 3}$$

$$\boxed{\frac{x}{y} = 3} \Rightarrow \underline{x = 3y}$$

then:

$$\begin{cases} x = 3y \\ x^2 - y^2 = 72 \end{cases}$$

$$\Rightarrow (3y)^2 - y^2 = 72$$

$$9y^2 - y^2 = 72$$

$$8y^2 = 72$$

$$y^2 = \frac{72}{8} = 9 \Rightarrow y = \sqrt{9}$$

$$\boxed{y = 3}$$

$$\Rightarrow x = 3(3)$$

$$\boxed{x = 9}$$

$$\Rightarrow 0 < 3 < 9 \text{ acc}$$

$$S: \{(9, 3)\}$$

E-K